

Greening Supply Chains or Muddying Waters? Investor Responses to the EU Deforestation Regulation

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Abstract

The EU Deforestation Regulation (EUDR) mandates that agrifood commodities sold in the EU derive from deforestation-free supply chains. In this paper, We study how global equity markets responded to the EUDR's regulatory trajectory using a Fama French five-factor event study on a sample of 2,031 listed agrifood firms across 40 countries. Adoption events in May and June 2023 generated near-zero average cumulative abnormal returns, consistent with a regulation that was largely anticipated by the time it passed. Postponement events tell a more nuanced story. The Commission's first delay proposal in October 2024 initially produced a positive market reaction, consistent with short-term compliance cost relief. However, adopting a rolling estimation window so that each event's CAR captures the marginal reaction net of what markets had already priced the subsequent Parliament vote produced a negative return (-0.67 pp), and both events of the second postponement cycle in 2025 were negative. The pattern suggests that while the first delay announcement was welcomed, continued regulatory uncertainty eroded value. We further analyse the effects by ESG grade heterogeneity and show that lower ESG firms outperformed at postponement events while the gap narrowed at adoption events. Our results contribute to the literature on regulatory credibility and asset prices, showing that markets can punish regulatory delay even when the delay temporarily reduces compliance costs.

Keywords: European Union, Sustainable Finance, Regulatory Governance, Environmental Governance, Comparative Political Economy

JEL Codes: G18, Q58, F53, P16

1 Introduction

Table 1: Event Timeline: EU Regulatory Announcements

Event	Description	Date
Event 1	EU Deforestation Regulation adopted	31 May 2023
Event 2	Regulation in Force	June 2023
Event 3	First postponement Proposal	October 2024
Event 4	First postponement Vote	Nov 2024
Event 5	Second postponement Proposal	October 2025
Event 6	Second postponement Vote	November 2025

Event	Mean CAR (percentage points)		
	CAR[−1, +1]	CAR[−3, +3]	CAR[−5, +5]
Adopted (May '23)	0.332** (0.132)	0.172 (0.196)	0.096 (0.249)
<i>N</i>	1437	1359	1355
In force (Jun '23)	-0.226* (0.130)	0.192 (0.179)	-0.153 (0.231)
<i>N</i>	1392	1387	1386
Delay P1 (Oct '24)	1.518*** (0.209)	0.923*** (0.265)	1.137*** (0.300)
<i>N</i>	1640	1562	1560
Delay V1 (Nov '24)	-0.673*** (0.157)	-1.813*** (0.223)	-1.520*** (0.270)
<i>N</i>	1594	1582	1573
Delay P2 (Oct '25)	-0.076 (0.129)	-0.788*** (0.189)	-0.593** (0.245)
<i>N</i>	1745	1638	1634
Delay V2 (Nov '25)	-0.750*** (0.120)	-2.136*** (0.184)	-2.679*** (0.236)
<i>N</i>	1743	1642	1641

Note:

Mean cumulative abnormal return (CAR) across all firms retained for each event (firms with fewer than 100 valid estimation-window days excluded). Standard errors computed as $SD/\sqrt{N}$. *N* = number of firms contributing to each CAR estimate. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-sided *t*-test against zero). Rolling 250-day pre-event window (ending at $t - 11$) per Bayer, Crippa & Genovese (2025).

Predictor	LASSO coefficient		
	CAR[−1, +1]	CAR[−3, +3]	CAR[−5, +5]
The Stock Exchange of Hong Kong Ltd		2.032	
Bolsa de Comercio de Buenos Aires	1.331		
China	1.105	1.571	
Brazil	0.923		
India		-0.725	
Korea; Republic (S. Korea)	-0.724		
Shenzhen Stock Exchange	0.678		
New Zealand		0.636	
Malaysia	-0.169		
Japan	0.088		
Canada	0.084		
Norway		0.063	
ESG grade B	0.063		
Thailand		-0.008	
Bolsa de Valores do Estado de Sao Paulo	0.006		
New Zealand Stock Exchange		0.002	
Tokyo Stock Exchange	0.001		
The Toronto Stock Exchange	0.001		
Korea Stock Exchange	-0.000		

Note:

$n = 1392$ firm–event observations (complete-case). Continuous predictors are z-scored; coefficients represent the change in CAR (pp) per one standard-deviation increase. Factor-level coefficients represent the change in CAR when that level is present vs. absent. Blank = predictor not selected for that window. Primary λ : lambda.min for CAR[−1, +1]; lambda.min for CAR[−3, +3]; lambda.min for CAR[−5, +5]. λ_{1se} falls back to λ_{min} when the CV curve is flat and the 1-SE tolerance includes the null model. All models use $\alpha = 1$ (LASSO), 10-fold cross-validation.

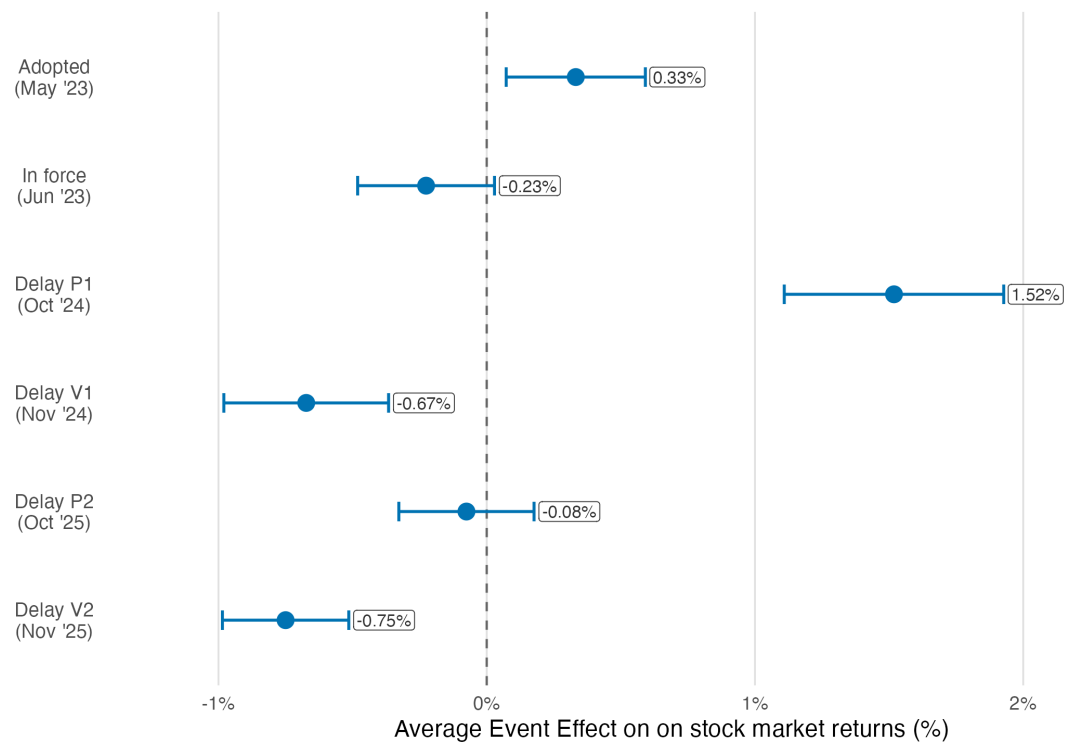
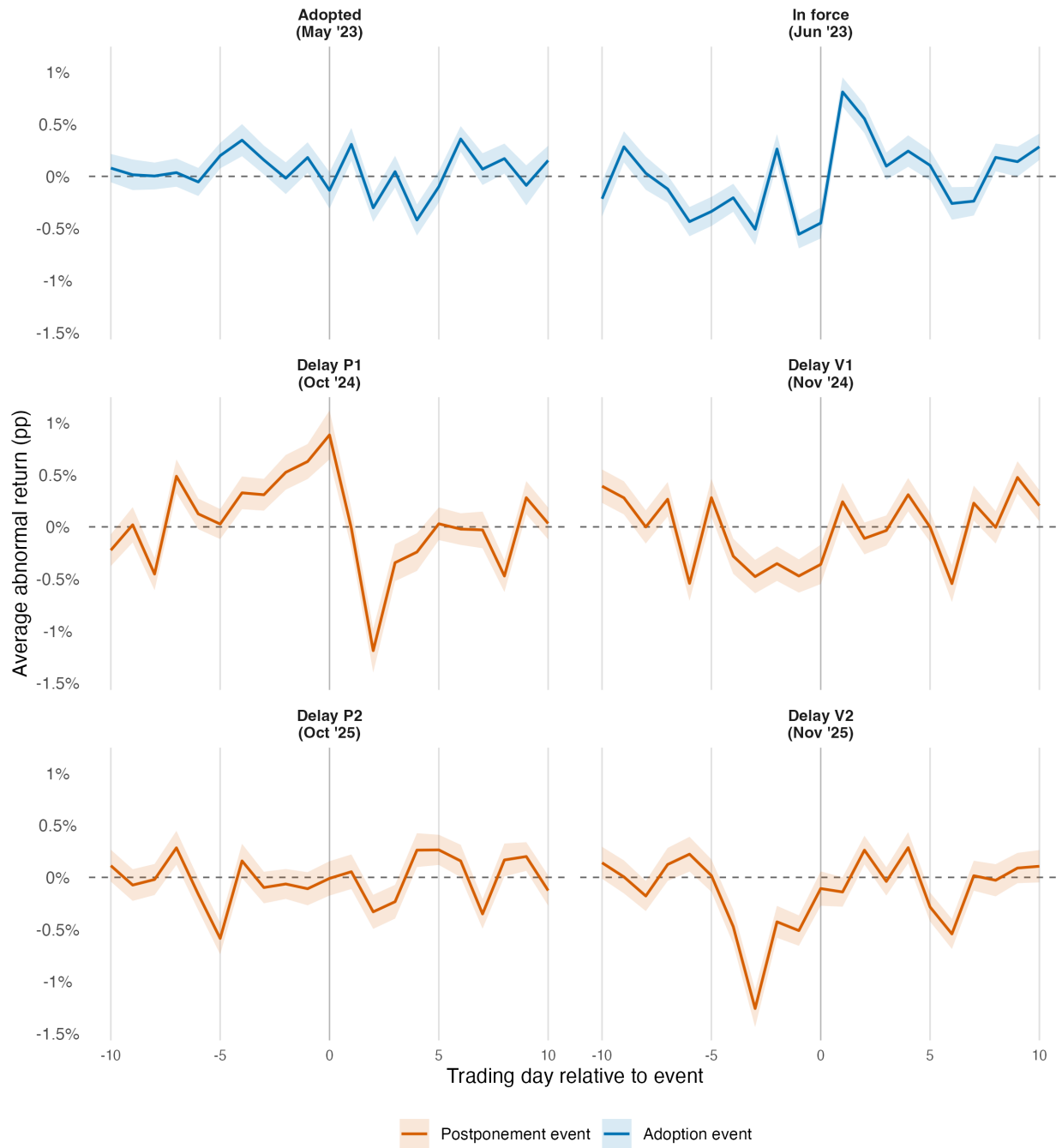


Figure 1: Local Average effect



Mean daily AR across all firms with non-missing returns for that event. Shaded ribbon: $\pm 1.96 \times \text{SE}$. Vertical line at $t=0$ (event day). Dashed line at zero.

Figure 2: Effect Dynamics

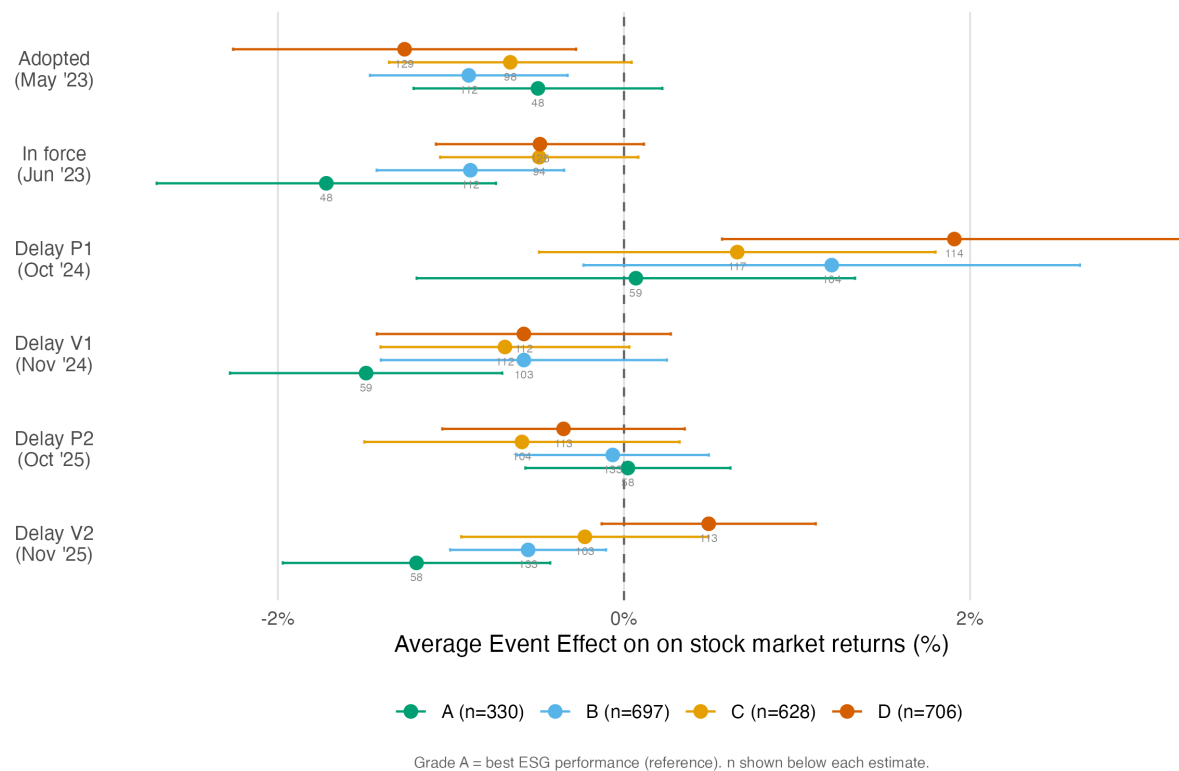


Figure 3: Effect by Environmental Pillar Score

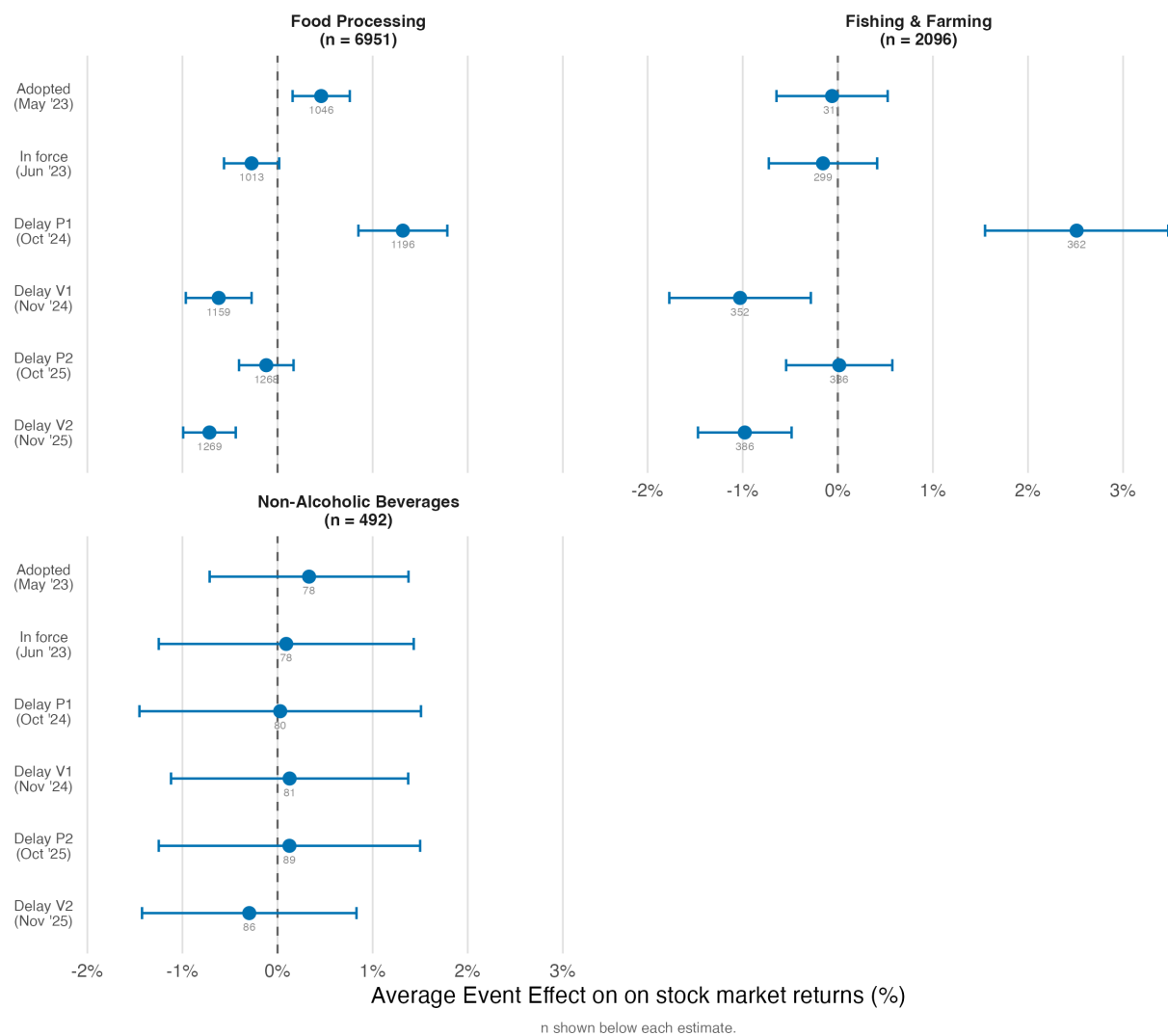


Figure 4: Effect by Industry